

Triggers

2520030575

M.LIKHITHA

Create a database & use:

```
1 CREATE DATABASE IF NOT EXISTS bookflow2_db;
2 • USE bookflow_db;
```

#	Time	Action
1	09:08:46	SHOW DATABASES
2	09:09:23	CREATE DATABASE IF NOT EXISTS bookflow2_db
3	09:09:40	USE bookflow_db

Create a table:

```
3 • CREATE TABLE Customer (
4     Customer_ID INT PRIMARY KEY,
5     Customer_Name VARCHAR(100) NOT NULL,
6     Phone VARCHAR(15),
7     Email VARCHAR(100),
8     City VARCHAR(50)
9 );
```

19	15:26:30	USE bookflow9_db
20	15:26:34	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) N...

```
10 • CREATE TABLE Account (
11     Account_No INT PRIMARY KEY,
12     Customer_ID INT,
13     Account_Type VARCHAR(20),
14     Balance DECIMAL(12,2) DEFAULT 0,
15     Branch VARCHAR(50),
16
17     FOREIGN KEY (Customer_ID)
18     REFERENCES Customer(Customer_ID)
19 );
```

#	Time	Action
20	15:26:34	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100)
21	15:27:34	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Typ

```

20 CREATE TABLE Bank_Transaction (
21     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
22     Account_No INT,
23     Transaction_Type VARCHAR(20),
24     Amount DECIMAL(12,2),
25     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
26
27     FOREIGN KEY (Account_No)
28     REFERENCES Account(Account_No)
29 );

```

Output

Action Output

#	Time	Action
21	15:27:34	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type ...
22	15:28:32	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Ac...

```

30 CREATE TABLE Loan (
31     Loan_ID INT PRIMARY KEY,
32     Customer_ID INT,
33     Loan_Type VARCHAR(30),
34     Loan_Amount DECIMAL(12,2),
35     Interest_Rate DECIMAL(5,2),
36
37     FOREIGN KEY (Customer_ID)
38     REFERENCES Customer(Customer_ID)
39 );

```

Output

Action Output

#	Time	Action
22	15:28:32	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Ac
23	15:29:04	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR

Insert data into customer table:

```

40 INSERT INTO Customer
41 (Customer_ID, Customer_Name, Phone, Email, City)
42 VALUES
43 (101, 'Lakshitha', '9876543210', 'ravi@gmail.com', 'Hyderabad'),
44 (102, 'Tejaswini', '9876543211', 'priya@gmail.com', 'Vijayawada'),
45 (103, 'Pranavi', '9876543212', 'arjun@gmail.com', 'Bangalore'),
46 (104, 'Suresh', '9876543213', 'sneha@gmail.com', 'Chennai'),
47 (105, 'Pounika', '9876543214', 'kiran@gmail.com', 'Hyderabad'),
48 (106, 'Sushma', '9876543215', 'anil@gmail.com', 'Delhi'),
49 (107, 'Navya', '9876543216', 'meena@gmail.com', 'Mumbai'),
50 (108, 'Aishwarya', '9876543217', 'rahul@gmail.com', 'Pune'),
51 (109, 'Vaishnavi', '9876543218', 'lakshmi@gmail.com', 'Hyderabad'),
52 (110, 'Aahila', '9876543219', 'suresh@gmail.com', 'Vijayawada');

```

Output

Action Output

#	Time	Action
54	17:19:02	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR
55	17:20:59	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Lakshitha', '98...

Insert into account:

```

53 INSERT INTO Account
54 (Account_No, Customer_ID, Account_Type, Balance, Branch)
55 VALUES
56 (10001, 101, 'Savings', 50000, 'Hyderabad'),
57 (10002, 102, 'Savings', 75000, 'Vijayawada'),
58 (10003, 103, 'Current', 120000, 'Bangalore'),
59 (10004, 104, 'Savings', 45000, 'Chennai'),
60 (10005, 105, 'Current', 90000, 'Hyderabad'),
61 (10006, 106, 'Savings', 63000, 'Delhi'),
62 (10007, 107, 'Current', 150000, 'Mumbai'),
63 (10008, 108, 'Savings', 35000, 'Pune'),
64 (10009, 109, 'Savings', 83000, 'Hyderabad'),
65 (10010, 110, 'Current', 110000, 'Vijayawada');

```

Output

Action Output

#	Time	Action
55	17:20:59	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Lakshitha', '98...
56	17:22:38	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, ...

Insert into bank transaction:

```
66 • INSERT INTO Bank_Transaction
67 (Account_No, Transaction_Type, Amount)
68 VALUES
69 (10001, 'DEPOSIT', 10000),
70 (10001, 'WITHDRAW', 5000),
71 (10002, 'DEPOSIT', 15000),
72 (10003, 'WITHDRAW', 20000),
73 (10004, 'DEPOSIT', 5000),
74 (10005, 'WITHDRAW', 10000),
75 (10006, 'DEPOSIT', 12000),
76 (10007, 'DEPOSIT', 25000),
77 (10008, 'WITHDRAW', 5000),
78 (10009, 'DEPOSIT', 20000),
79 (10010, 'WITHDRAW', 15000);
```

Output

Action Output

#	Time	Action
86	17:22:08	INSERT INTO Account (Account_No, Customer_ID, Account_Type
87	17:22:39	INSERT INTO Bank_Transaction (Account_No, Transaction_Type,

Insert into loan:

```
80 • INSERT INTO Loan
81 (Loan_ID, Customer_ID, Loan_Type,
82 Loan_Amount, Interest_Rate)
83 VALUES
84 (501, 101, 'Home Loan', 5000000, 7.5),
85 (502, 102, 'Education Loan', 1000000, 6.5),
86 (503, 103, 'Car Loan', 800000, 8.2),
87 (504, 104, 'Personal Loan', 500000, 10.5),
88 (505, 105, 'Home Loan', 4000000, 7.2),
89 (506, 106, 'Car Loan', 900000, 8.5),
90 (507, 107, 'Business Loan', 3000000, 9.0),
91 (508, 109, 'Personal Loan', 600000, 10.0);
```

Output

Action Output

#	Time	Action
87	17:22:39	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount
88	17:23:14	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, I

Display all tables:

```
92 • SELECT * FROM Customer;
```

Result Grid

	Customer_ID	Customer_Name	Phone	Email	City
▶	101	Lidhitha	9876543210	ravi@gmail.com	Hyderabad
	102	Tejaswini	9876543211	priya@gmail.com	Vijayawada
	103	Pranavi	9876543212	arjun@gmail.com	Bangalore
	104	Ganesh	9876543213	sneha@gmail.com	Chennai
	105	Mounika	9876543214	kiran@gmail.com	Hyderabad
	106	Sushma	9876543215	anil@gmail.com	Delhi
	107	Navya	9876543216	meena@gmail.com	Mumbai
	108	Aishwarya	9876543217	rahul@gmail.com	Pune
	109	Vaishnavi	9876543218	lakshmi@gmail.com	Hyderabad
	110	Akhila	9876543219	suresh@gmail.com	Vijayawada

93 • `SELECT * FROM Account;`

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
10010	110	Current	110000.00	Vijayawada
NULL	NULL	NULL	NULL	NULL

94 • `SELECT * FROM Bank_Transaction;`

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-16 17:22:39
2	10001	WITHDRAW	5000.00	2026-08-16 17:22:39
3	10002	DEPOSIT	15000.00	2026-08-16 17:22:39
4	10003	WITHDRAW	20000.00	2026-08-16 17:22:39
5	10004	DEPOSIT	5000.00	2026-08-16 17:22:39
6	10005	WITHDRAW	10000.00	2026-08-16 17:22:39
7	10006	DEPOSIT	12000.00	2026-08-16 17:22:39
8	10007	DEPOSIT	25000.00	2026-08-16 17:22:39
9	10008	WITHDRAW	5000.00	2026-08-16 17:22:39
10	10009	DEPOSIT	20000.00	2026-08-16 17:22:39
11	10010	WITHDRAW	15000.00	2026-08-16 17:22:39
NULL	NULL	NULL	NULL	NULL

95 • `SELECT * FROM Loan;`

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Intere
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	4000000.00	7.20
506	106	Car Loan	900000.00	8.50
507	107	Business Loan	3000000.00	9.00
508	109	Personal Loan	600000.00	10.00
NULL	NULL	NULL	NULL	NULL

Create a view for all customers:


```

96 • CREATE VIEW Customer_View AS
97 SELECT *
98 FROM Customer;
99 • SELECT * FROM Customer_View;

```

Result Grid | Filter Rows: | Exports: | Wrap Cell Contents:

Customer_ID	Customer_Name	Phone	Email	City
101	Likhitha	9876543210	ravi@gmail.com	Hyderabad
102	Tejaswini	9876543211	priya@gmail.com	Vijayawada
103	Pranavi	9876543212	arjun@gmail.com	Bangalore
104	Ganesh	9876543213	sneha@gmail.com	Chennai
105	Mounika	9876543214	kiran@gmail.com	Hyderabad
106	Sushma	9876543215	anil@gmail.com	Delhi
107	Navya	9876543216	meena@gmail.com	Mumbai
108	Aishwarya	9876543217	rahu@gmail.com	Pune
109	Vaishnavi	9876543218	lakshmi@gmail.com	Hyderabad
110	Akhila	9876543219	suresh@gmail.com	Vijayawada

Create a view for customer basic information:

```

100 • CREATE VIEW Customer_Basic_View AS
101 SELECT
102     Customer_ID,
103     Customer_Name,
104     City
105 FROM Customer;
106 • SELECT * FROM Customer_Basic_View;

```

Result Grid | Filter Rows: | Exports:

Customer_ID	Customer_Name	City
101	Likhitha	Hyderabad
102	Tejaswini	Vijayawada
103	Pranavi	Bangalore
104	Ganesh	Chennai
105	Mounika	Hyderabad
106	Sushma	Delhi
107	Navya	Mumbai
108	Aishwarya	Pune
109	Vaishnavi	Hyderabad
110	Akhila	Vijayawada

Create a view for account information:

```

• CREATE VIEW Account_View AS
SELECT
    Account_No,
    Account_Type,
    Balance,
    Branch
FROM Account;
• SELECT * FROM Account_View;

```

Account_No	Account_Type	Balance	Branch
0001	Savings	50000.00	Hyderabad
0002	Savings	75000.00	Vijayawada
0003	Current	120000.00	Bangalore
0004	Savings	45000.00	Chennai
0005	Current	90000.00	Hyderabad
0006	Savings	65000.00	Delhi
0007	Current	150000.00	Mumbai
0008	Savings	35000.00	Pune
0009	Savings	85000.00	Hyderabad
0010	Current	110000.00	Vijayawada

Create a view for savings account:

```

115 • CREATE VIEW Savings_Account_View AS
116     SELECT *
117     FROM Account
118     WHERE Account_Type = 'Savings';
119 • SELECT * FROM Savings_Account_View;

```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10004	104	Savings	45000.00	Chennai
10006	106	Savings	65000.00	Delhi
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad

High balance account view:

```

• CREATE VIEW High_Balance_View AS
SELECT
    Account_No,
    Customer_ID,
    Account_Type,
    Balance
FROM Account
WHERE Balance > 100000;
• SELECT * FROM High_Balance_View;

```

Account_No	Customer_ID	Account_Type	Balance
10003	103	Current	120000.00
10007	107	Current	150000.00
10010	110	Current	110000.00

Customer + account view:

```

130 SELECT
131     C.Customer_ID,
132     C.Customer_Name,
133     C.City,
134     A.Account_No,
135     A.Account_Type,
136     A.Balance,
137     A.Branch
138 FROM Customer C
139 JOIN Account A
140 ON C.Customer_ID = A.Customer_ID;
141 * SELECT * FROM Customer_Account_View;

```

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
101	Likhitha	Hyderabad	10001	Savings	50000.00	Hyderabad
102	Tejaswini	Vijayawada	10002	Savings	75000.00	Vijayawada
103	Pranavi	Bangalore	10003	Current	120000.00	Bangalore
104	Ganesh	Chennai	10004	Savings	45000.00	Chennai
105	Mounika	Hyderabad	10005	Current	90000.00	Hyderabad
106	Sushma	Delhi	10006	Savings	65000.00	Delhi
107	Navya	Mumbai	10007	Current	150000.00	Mumbai
108	Aishwarya	Pune	10008	Savings	35000.00	Pune
109	Vaishnavi	Hyderabad	10009	Savings	85000.00	Hyderabad
110	Akhila	Vijayawada	10010	Current	110000.00	Vijayawada

Complete customer bank view:

```

146     C.City,
147     A.Account_No,
148     A.Account_Type,
149     A.Balance,
150     A.Branch,
151     L.Loan_Type,
152     L.Loan_Amount
153 FROM Customer C
154 LEFT JOIN Account A
155 ON C.Customer_ID = A.Customer_ID
156 LEFT JOIN Loan L
157 ON C.Customer_ID = L.Customer_ID;
158 * SELECT * FROM Customer_Banking_View;

```

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch	Loan_Type	Loan_Amount
101	Likhitha	Hyderabad	10001	Savings	50000.00	Hyderabad	Home Loan	5000000.00
102	Tejaswini	Vijayawada	10002	Savings	75000.00	Vijayawada	Education Loan	1000000.00
103	Pranavi	Bangalore	10003	Current	120000.00	Bangalore	Car Loan	800000.00
104	Ganesh	Chennai	10004	Savings	45000.00	Chennai	Personal Loan	500000.00
105	Mounika	Hyderabad	10005	Current	90000.00	Hyderabad	Home Loan	4000000.00
106	Sushma	Delhi	10006	Savings	65000.00	Delhi	Car Loan	900000.00
107	Navya	Mumbai	10007	Current	150000.00	Mumbai	Business Loan	3000000.00
108	Aishwarya	Pune	10008	Savings	35000.00	Pune	NA	NA
109	Vaishnavi	Hyderabad	10009	Savings	85000.00	Hyderabad	Personal Loan	600000.00
110	Akhila	Vijayawada	10010	Current	110000.00	Vijayawada	NA	NA